

Mohan Raj Senthilnathan

Principal Software Engineer | Gen (NortonLifeLock / Symantec)

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Profile Summary

Innovative Principal Software Engineer with 12+ years of experience in architecting scalable, secure, and high-performance distributed systems. Proven expert in modernizing legacy architectures and driving end-to-end delivery of complex, multi-tenant cloud-native platforms.

Pioneer in AI-driven Engineering: Designed and deployed autonomous AI agents using the Claude Agent SDK and AWS Bedrock to automate test triage, summarize sprint activities, and remediate up to 60% of regression failures without human intervention.

Technical Skills

- Languages: C#, Java, Python, JavaScript, Scala (Beginner)
- AI & Agents: **Claude Agent SDK, AWS Bedrock**, LangChain, LangGraph, Prompt Engineering, LLM Integration, RAG.
- Frameworks: .NET Core, ASP.NET, Django, Angular, React, Node.js
- Databases: PostgreSQL (Relational), Azure Cosmos DB (NoSQL)
- Cloud & DevOps: Azure Services, Azure DevOps, GCP, Kubernetes, Docker, Nginx, Redis(caching)
- Messaging & Streaming: Kafka, RabbitMQ
- Distributed Systems: Microservices Architecture, Event-Driven Systems, Fault Tolerance, Distributed Tracing, Scalability Patterns
- Tools & Technologies: RESTful APIs, GraphQL, OpenID Connect, Webpack, Adobe Target
- Observability: Prometheus (Metrics), Grafana (Dashboards), Datadog, Sumo Logic, Splunk
- Core Expertise: System Design, Scalable Architecture, Performance Tuning, TDD, Design Patterns, Agile/Scrum
- Security: OAuth 2.0, JWT, OWASP Practices
- Principles: SOLID, DRY, KISS, YAGNI

Professional Experience

Gen (NortonLifeLock / Symantec) – Chennai (9+ Years)

- Principal Software Engineer | Jul 2023 – Present
- Senior Software Engineer | Aug 2020 – Jul 2023
- Software Engineer | Jan 2017 – Aug 2020

Key Contributions in MultiTenant Web Portal (Since Jan 2017):

- **Designed and developed a multi-tenant portal from scratch** — an extensible platform for managing subscriptions, sharing licenses, managing devices, and handling automatic renewals. The portal was extended for use across Norton, Avast, and Avira.
- **Built an AI-powered test result classification system** using Claude Agent SDK to intelligently triage test outcomes, reducing manual analysis effort and accelerating defect identification in CI/CD pipelines. (AI)
- **Developed an AI-powered tool using AWS Bedrock** to summarize discussions and automatically generate Jira stories/defects from test cases, eliminating manual ticket creation and improving team efficiency (AI)
- **Engineered an automated defect remediation framework** capable of resolving up to 60% of regression failures without human intervention, significantly minimizing debugging time. (AI)
- **Leveraged Claude Agent SDK to analyze sprint check-ins** and identify regression hotspots, enabling targeted testing efforts and improving defect detection efficiency through AI-driven insights. (AI)
- **Architected a robust security layer** for the Gen portal using **OpenID Connect (OIDC)** and **RESTful APIs** to ensure secure, standardized authentication and authorization across distributed multi-tenant services
- **Optimized Cosmos DB queries** by using selective fields and custom indexing, reducing RU consumption and improving performance
- **Integrated Redis cache to synchronize data across servers**, reducing backend calls, improving performance, and ensuring consistency in a distributed environment.

Technologies used in MultiTenant Web Portal:

.NET, C#, JavaScript, Python, react, Azure, Team City, Redis, Cursor CLI, Azure Cosmos DB, React.JS.

Key Contributions in Email Guardian (Since Oct 2024):

- **Designed and implemented Feedback Loop enhancements in Email Guardian**, enabling customers to report incorrectly scanned emails and triggering automatic re-scans; developed the complete data flow using a Pub/Sub model.
- **Designed a high-throughput Metrics API** utilizing Kafka and GCP BigQuery to enable real-time scalable data collection and advanced analytics.
- **Built an internal Web Application tool for customer support**, allowing support engineers to efficiently manage and resolve queries related to onboarded emails, improving overall team productivity.
- **Developed a microservice orchestrator using RabbitMQ**, enhancing backend communication efficiency and enabling scalable, reliable, message-driven workflows.
- **Executed cost-efficiency initiatives** by migrating Nginx to a free tier through custom open-source authentication modules, reducing licensing overhead.
- **Implemented Database Sharding** to enhance horizontal scalability, successfully facilitating the seamless onboarding of the Norton tenant onto the platform.

Technologies used in Email Guardian:

Java, Spring Boot, Kubernetes, Docker, Kafka, PostgreSQL, Nginx, GCP, Team city, Grafana, Python.

Other Contributions in Current Company (Leadership Role):

- **Mentored and guided junior engineers**, fostering skill development and improving team productivity.
- **Championed knowledge-sharing sessions, technical workshops**, and design discussions to uplift team competency.
- **Led epic estimation and dependency mapping**, ensuring cross-team alignment and proactive resolution of technical blockers.
- **Led code reviews and promoted development best practices**, ensuring high-quality, maintainable software.
- **Led AI tool adoption**, including prompt engineering, rule creation, and specification design, improving team efficiency and capabilities. (AI)
- **Developed open-source C# and JS libraries**, promoting code reuse and community engagement.

HCL Technologies (Client: Cisco Systems) – Chennai

- Member Technical Staff | Mar 2015 – Dec 2016

Key Contributions in Cisco Identity Services Engine:

- Developed REST APIs and backend services for Cisco Identity Services Engine (ISE) platform.
- Built internal developer productivity tools improving turnaround time for development and QA teams.

Amazon Development Center – Chennai

- Device Associate | Oct 2013 – Mar 2015

Key Contributions in Project B (Amazon Fire Phone):

- Owned module-level testing for Amazon Fire Phone & Kindle devices, automating key test cases to enhance release quality.

Education

B. Tech in Information Technology | Anna University | 2009 – 2013

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) – 2022

Patent

Systems and Methods for Restricting Capture of Self-Portrait Images on Mobile Devices

Patent No: 11528355 | Issued: Dec 13, 2022